```python
import tensorflow as tf
import numpy as np

# Generate some random data
np.random.seed(0)
X = np.random.rand(1000, 2)
y = np.random.randint(0, 2, size=1000)

# Split the data into training and testing sets
x_train, x_test = X[:800], X[800:]
y_train, y_test = y[:800], y[800:]

# Define the logistic regression model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(1, activation='sigmoid', input_dim=2)
])

# Compile and train the model
model.compile(optimizer='adam',
              loss=tf.keras.losses.BinaryCrossentropy(from_logits=False),
              metrics=['accuracy'])

model.fit(x_train, y_train, epochs=10)

# Evaluate the model on the testing data
loss, accuracy = model.evaluate(x_test, y_test)
print('Test loss:', loss)
print('Test accuracy:', accuracy)
```

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` argumer
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
25/25 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.5699 - loss: 0.6883
Epoch 2/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5123 - loss: 0.7067
Epoch 3/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5137 - loss: 0.7017
Epoch 4/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.5052 - loss: 0.7018
Epoch 5/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5324 - loss: 0.6942
Epoch 6/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5343 - loss: 0.6939
Epoch 7/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5437 - loss: 0.6931
Epoch 8/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5341 - loss: 0.6938
Epoch 9/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5587 - loss: 0.6917
Epoch 10/10
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.5071 - loss: 0.6959
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.4889 - loss: 0.6956
Test loss: 0.6914459466934204
Test accuracy: 0.5099999904632568
```